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PHY 311 — Past Questions & Materials
by Renance

Course: PHY 311
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 15
Levels: 300

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Paper 1 of 15 — PHY 311 - 2022/2023 - First Semester - Quantum Mechanics I

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 311

Course code: Quantum Mechanics I

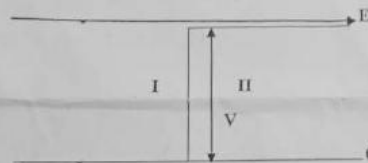
Time allowed: 3 hrs

Unit: 3 units

Date: 26/02/2021: 8.00 AM

Instructions: Answer any four questions

- (1a). List and explain with equations, the three pictures of quantum mechanics. (7½ marks)
- (1b). Show that $\psi = e^{-i\omega t/2}$ satisfies the Schrodinger wave equation with $H = \omega S_z$. (4 marks)
- (1c). Obtain the SWE for a charged particle in an electromagnetic field characterized by the vector potential $\vec{A}$, and the scalar potential ϕ whose classical Hamiltonian is given by (6 marks)
- $$H = \frac{1}{2m} (\vec{p} - q\vec{A})^2 + q\phi$$
- (2). Consider a particle of mass m , and Energy E in the following potential step $V(x)$



$$V(x) = V_1 \quad \forall \quad -\infty < x < 0$$

$$= V_2 \quad \forall \quad 0 < x < \infty$$

Where I is the incident region and II is the transmitted region (i). Find the amplitudes at the boundary $x=0$ when $E > V$ (8½ marks) (ii). Find the reflection and transmission coefficients when $E > V$ (9 marks) Using the time independent Schrodinger wave equation.

$$\left[\frac{-\hbar^2 d^2}{2mdx^2} + V(x) \right] \psi(x) = E\psi(x)$$

- (3). A particle of mass m moves in a potential given by $V(x, y, z) = \frac{k}{2} (x^2 + y^2 + z^2) = \frac{kr^2}{2}$
- (a). Write down the time-independent Schrodinger equation for this system. (1 mark)
- (b). Make the substitution $\psi(x, y, z) = X(x)Y(y)Z(z)$ and separate the variables for this system (5 marks)
- (c). What are the solutions to the resulting equations for $X(x)Y(y)Z(z)$ (2½ marks)
- (d). What is the general expression for the quantized energy/levels of this system, in terms of the quantum number n_x, n_y and n_z which corresponds to $X(x)Y(y)Z(z)$ (1 mark)
- (e). What is the degree of degeneracy of a state of energy $E = 5.5\hbar \sqrt{\frac{k}{m}}$ for this system? (4 marks)
- (f). An alternative solution may be the substitution, what are the solutions for $\psi(r, \theta, \phi) = F(r) G(\theta, \phi)$. In this substitution, what are the solutions for $G(\theta, \phi)$. (1 mark)
- (g). Write down the differential equation for $F(r)$ which is obtained when the substitution $\psi(r, \theta, \phi) = F(r) G(\theta, \phi)$ is made. Do not solve this equation. (3 marks)

- (4). For a 1-D Simple harmonic oscillator where $f(x) = -kx$, $V = \frac{1}{2}kx^2$. If the One-dimensional Schrodinger time-independent wave equation for the problem is given by $\left[\frac{-\hbar^2}{2m}\frac{d^2}{dx^2} + V\right]\psi = E\psi$ with the transformations $k = m\omega^2$, $y = \left(\frac{m\omega}{\hbar}\right)^{1/2}x = \beta x$, $\lambda = \frac{2E}{\hbar\omega}$, assuming a solution $\psi = H(y)e^{-1/2y^2}$, derive the ground state energy $E = \frac{\hbar\omega}{2}(2N+1)$ (17 $\frac{1}{2}$ marks)
- (5a). List five characteristic properties of the wave function of the quantum theory. (5 marks)
- (5b). Normalize the following wave functions. (i). $\psi = xe^{-\alpha x^2}$ (2 marks)
- (ii). $\psi = Ax^2$ where $-b/2 < x < b/2$ (2 marks)
- (iii). $\psi(x) = A\sqrt{x} \cos\left(\frac{\pi x}{2b}\right)$ $0 < x < b$ (4 $\frac{1}{2}$ marks)
- (iv). $\psi = Ax(a-x)$ $0 \leq x < a$; $0 \leq x \leq a/b$ (4 marks)
- (6a). (i). Define an operator in the Hilbert space and in linear algebra. (1 $\frac{1}{2}$ marks)
- (ii). Write a short note on Hermitian operator and Hermitian matrix. (1 $\frac{1}{2}$ marks)
- (iii). Write the relation and the general solution for the Schrodinger equation in spherical coordinates. (1 $\frac{1}{2}$ marks)
- (iv). Write a short note on expectation values. (1 $\frac{1}{2}$ marks)
- (6b). The normalized function of a particle is given by $\int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$, if the particle is found between $0 \leq x \leq L$, find the expectation value of the momentum P_x and position X . (6 $\frac{1}{2}$ marks)
- (6c). Obtain the expectation value for the momentum of a particle whose wave function is given by $\psi_1(x) = \frac{1}{\sqrt{b}} \exp(ikx)$ if $0 \leq x \leq b$ (5 marks)

Time Independent $\rightarrow i\hbar \frac{\partial^2}{\partial x^2} \psi(x) + V(x) \psi(x) = E \psi(x)$

Time Independent $\rightarrow i\hbar \frac{\partial}{\partial t} \psi(x,t) + V(x) \psi(x,t) = i\hbar \frac{\partial}{\partial t} \psi(x,t)$

Paper 2 of 15 — PHY 311 - 2017/2018 - First Semester - Quantum Mechanics

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS
FIRST SEMESTER EXAMINATION

Academic Session: 2017/2018
Course Code: PHY 311
Course Title: Quantum Mechanics
Course Unit: 3
Time Allowed: 2hrs 30mins
Instruction: Answer any Four Questions

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- 1a) Obtain the Schrodinger wave equation for a charged particle in an electromagnetic field characterized by the vector potential $\vec{A}$ and the scalar potential ϕ whose classical Hamiltonian is given by

$$H = \frac{1}{2m} (\vec{p} - q\vec{A})^2 + q\phi$$

- 1b) The Hamiltonian of an electron situated in a magnetic field of induction $\vec{B} = (0, 0, B)$, is given by $H = \frac{e}{m} \vec{B} \cdot \vec{S}$. Obtain the equation of motion for $\langle S_x \rangle$, $\langle S_y \rangle$ and $\langle S_z \rangle$ and comment on your results.

- 1c) Show that $[L_x, L_y] = i\hbar L_z$

- 1d) If the energy difference in an infinitely deep potential well is $E_0 = 5J$ and $E_1 = 12J$, find the width of the potential well.

- 2) For a particle scattered from a potential step, obtain the relations for the reflection and transmission coefficients when $E > 0$

- 3a) A particle moving along the x-axis within the limits $(0, a)$ is described by the wavefunction

$$\psi = \sin \frac{\pi x}{a}$$

- i) Determine the normalizing factor and mean position of the particle.
ii) Determine the uncertainty Δx in the position of the particle.

- 3b) A particle trapped in a region may be described by the wavefunction $\psi = Ax(a-x)$ within the interval; $0 \leq x \leq a = 0$. Where A is the normalizing factor. Determine A, and the probability of finding the particle within the interval $0, \frac{a}{3}$

- 3c) Show that $\psi = \begin{pmatrix} e^{-w/2} \\ e^{w/2} \end{pmatrix}$ satisfies the Schrodinger wave equation with $H = wS_z$.

- 4a) List and describe the three (3) pictures of quantum mechanics.

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4b) For $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$; with $0 < x < L$, find the average value of $P_x \cdot X$. That is $\langle P_x \cdot x \rangle$

4c) Evaluate $\langle \psi | H' | \psi \rangle$ for a simple harmonic oscillator, where $\psi = \left(\frac{mw}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{mw}{2\hbar} x^2\right)$ and $H' = ve^{-\beta x^2}$

4d) A system is in a state described by the wave function $\psi = \frac{K_1^{1/2}}{\pi^{1/4}} \exp\left(-\frac{1}{2} K_1^2 x^2\right)$. Determine the probability that a measurement of its energy will give a result corresponding to the energy Eigenfunction $\phi_0 = \frac{K_1^{1/2}}{\pi^{1/4}} \exp\left(-\frac{1}{2} K_1^2 x^2\right)$

5a) A particle confined to a cubical box of volume V is found to be in a state given by the wavefunction $|\psi\rangle = Axyz$

- Obtain the normalizing constant.
- Determine the probability of finding the particle in the ground state of the box.

5b) For a particle scattered from a potential step at $E > V$, find the percentage transmission and reflection if

- $E = 6eV$, $V = 2eV$
- $E = 3.24eV$, $V = 2eV$
- Determine $x = \frac{E}{V}$ for $T = 75\%$

5c) Normalize the wavefunction $|\psi\rangle = \begin{pmatrix} 2-i \\ -3 \\ 2+3i \end{pmatrix}$

6a) The transmission coefficient in the scattering from a potential barrier is given as

$$T = \left[1 + \frac{V^2 \sin^2 k_1^2 a}{4E(E-V)}\right]^{-1} \text{ where } k_1^2 = \frac{2m}{\hbar^2}(E-V)$$

- If the width of the potential barrier is $a = 0.280nm$, the energy and potential are $15eV$ and $12eV$ respectively, determine T
- If the energy and potential are $14eV$ and $16eV$ respectively for a reflection coefficient of $R = 90\%$, find k and a

6b) $\hat{A} = \begin{pmatrix} 1 & i\sqrt{3} & -i \\ -i\sqrt{3} & 0 & 3 \\ i & 3 & 1 \end{pmatrix}$, $\psi_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ i \\ 1 \end{pmatrix}$, $\psi_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

Determine the expectation value of $\hat{A}$ in each of the state ψ_1 and ψ_2 .

6c) Determine the average of x^2 for a simple harmonic oscillator which is in a state defined by the wavefunction ψ is e^{-ax^2} .

$$8.7450 \times 10^{-30} = 7.6634 \times 10^{-30}$$

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Paper 3 of 15 — PHY 311 - 2017/2018 - First Semester - Physics

Level 300 • First Semester • PastQuestion

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PHY 311 (1st term) 12/4/2018.
2017/2018 session. www.fupreblog.com

Time $\rightarrow$ 45 mins.

Instruction - Answer 2 questions $\int \cos x e^{-\sin x} dx$

1. If $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$ for $0 < x < L$
show that $\langle p_x \rangle = 0$ if $\langle p_x \rangle = \frac{\int \psi^* p_x \psi}{\int \psi^* \psi}$
all space

2) If $\psi = \sin \frac{\pi x}{9}$, $0 < x < 9$; (i) determine the normalisation factor and mean position of the particle
ii) Determine the uncertainty, Δx in the position of the particle

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Paper 4 of 15 — PHY 311 - 2016/2017 - First Semester - Quantum Mechanics

Level 300 • First Semester • PastQuestion

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COLLEGE OF SCIENCE

H DEPARTMENT OF PHYSICS

1 FIRST SEMESTER EXAMINATION

: Academic Session: 2016/2017 -

Course Code: PHY 311

Course Title: Quantum Mechanics

Course Unit: < Ie F

1 Time Allowed: 3 Hrs

=at- Instruction;7 Answer any Four Questions

Q)) Explain the Heisenberg uncertainty relation and Fourier decomposition

) If $|y(x)|^2$ is the probability to find the particle within $x, x+dx$, obtain the relation for the uncertainty in the particle's position and the uncertainty in the momentum if the probability density for the momentum is $\phi(p)$

i (©) Prove that for any function $\psi(x)$, we have the inequality $\Delta x \Delta p \geq \hbar/2$

| @ What does Compton scattering show

3 (2) { @ Enumerate the five rules on how photon scattering changes the electron's wave

(b) Explain the following

| i (1) Thompson model

j (2) Rutherford scattering

. (3) Franck- Hertz experiment

i; (4) The Bohr atoms

| arn Roa

(b) List 3 postulates of quantum mechanics “

. (c) Show the continuity equation classically. Hence show that the wave function is normalized.

(@) Write out the operators for the following physical quantities (1) KE (2) Total energy (3) momentum (4) potential energy (4) $\langle p \rangle$ (5) Velocity $\langle v \rangle$

@} Gy Obtain the lowest energy for a particle in an infinitely deep rectangular well, where $0 < x < a$.

E is 15 joules and E_2 is 27 joules, what is the width of the well.

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} (@)List five axioms that defines a vector space

j | (S) For @ simple Harmonic oscillator, show that the ground state energy is $B = \hbar \omega / 2$

' (O)pf2f List and explain the three pictures of quantum mechanics

| <W) Solve the Schrodinger wave equation for a hydrogen atom in a spherical coordinate.

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Paper 5 of 15 — PHY 311 - 2012/2013 - First Semester - QUANTUM MECHANICS I

Level 300 • First Semester • PastQuestion

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EFFURUN, NIGERIA


 COLLEGE OF SCIENCE
 DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2012/2013 ACADEMIC SESSION
 COURSE CODE: PHY 311 TIME ALLOWED: 3 HRS
 COURSE TITLE: QUANTUM MECHANICS I COURSE UNIT: 3
 INSTRUCTION: ANSWER ANY FOUR QUESTIONS

1a. Write down the three postulates of Quantum mechanics

1b. Explain

1. The correspondence principle
2. The Heisenberg uncertainty principle
3. The complementarity principle

1c. What is the De Broglie wavelength of an electron whose kinetic energy is 100eV ($h = 6.626 \times 10^{-34}$ Js, $m_e = 9.109 \times 10^{-31}$ Kg).

2a. Using the uncertainty principle, estimate the ground state energy of an electron in the hydrogen atom.

2b. Obtain the general solution for a particle in an infinitely deep potential well.

3a. Prove that $[L_1, L_2] = i\hbar \sum_{123} L_3$ that is $[L_1, L_2] = i\hbar L_3$

3b. A particle trapped in a region may be described by the wave function $\psi = Ax(a-x)$ for $0 < x < a$. Determine A and the probability of finding the particle within the interval $(\frac{a}{3}, \frac{2a}{3})$.

4a. Starting from the time dependent Schrodinger wave equation, obtain the exact solution for a free particle.

4b. If $\psi(\vec{r}, t) = \phi(\vec{r})e^{-iEt/\hbar}$, prove that it satisfies the mass conservation equation $\nabla \cdot \vec{j} = 0$.

5a. For a potential barrier of infinite width, obtain the general solution for scattering from a potential step of height V_0 in the regions $x < 0$ and $x > 0$.

5b. If $\psi(x) = A\sqrt{x} \cos\left(\frac{\pi x}{2b}\right)$ in the region $0 < x < b$, find the constant of normalization.

6a. A particle in an infinite potential well $0 < x < a$ is described by the wavefunction $\psi = \sqrt{3}x$. Determine the probability that it is in the third excited state.

6b. Derive the relation $\frac{d^2\phi}{dy^2} + (\lambda - y^2)\phi = 0$ for the one dimensional simple harmonic oscillator.

(in $(\pi/2 \text{ side})^2$)

Paper 6 of 15 — PHY 311 - 2015/2016 - First Semester - QUANTUM MECHANICS I

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES
EFFURUN, NIGERIA
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2015/2016 ACADEMIC SESSION
COURSE CODE: PHY 311 TIME ALLOWED: 2:30 Hrs
COURSE TITLE: QUANTUM MECHANICS I COURSE UNIT: 3
INSTRUCTION: ANSWER ANY FOUR QUESTIONS

- 1a. List five characteristics of a wavefunction
- 1b. For a particle in the state given by $\psi = Ax^2$, $0 < x < a$, determine the normalization constant A and the probability that the measurement will yield an eigenvalue corresponding to the normalized eigenfunction, $\psi = \sqrt{\frac{2}{a}} \sin \frac{2\pi x}{a}$.
- 2a. What is the De Broglie wavelength of an electron whose kinetic energy is 100eV ($\hbar = 6.626 \times 10^{-34}$ Js, $m_e = 9.109 \times 10^{-31}$ Kg).
- 2a. Using the uncertainty principle, estimate the ground state energy of an electron in the hydrogen atom
- (b) Obtain the general solution for a particle in an infinitely deep potential well.
- 3a. Prove that $[L_x, L_y] = i\hbar L_z$ that is $[L_x, L_y] = i\hbar L_z$
- (b) Explain;
1. The correspondence principle
 2. The Heisenberg uncertainty principle
 3. The complementarity principle
- 4a. Using the time dependent Schrodinger wave equation, show that $\psi(\vec{r}, t) = \phi(\vec{r})e^{-\frac{iEt}{\hbar}}$
- 4b. If $\psi(\vec{r}, t) = \phi(\vec{r})e^{-\frac{iEt}{\hbar}}$ prove that it satisfies the mass conservation equation
- 5a. A system is in a state described by the wave function $\psi = \frac{\sigma^K}{\pi^K} \exp\left(-\frac{1}{2}\sigma^2 x^2\right)$. Determine the probability that a measurement of its energy will give a result corresponding to the energy eigenfunction $\phi_0 = \frac{\alpha^K}{\pi^K} \exp\left(-\frac{1}{2}\alpha^2 x^2\right)$
- 5b. An electron situated in a magnetic field of induction $\vec{B} = (0, 0, B)$ have a Hamiltonian given by $H = \frac{e}{m} \vec{B} \cdot \vec{S}$. Obtain the equation of motion for $\langle S_x \rangle$, $\langle S_y \rangle$ and $\langle S_z \rangle$
- 5c. List and describe adequately, the three pictures of quantum mechanics.
- 6a. Given that $A = (P - i\lambda x) \frac{1}{\sqrt{\gamma}}$ and $A^\dagger = (P + i\lambda x) \frac{1}{\sqrt{\gamma}}$ where λ, γ are constants and all other quantities have their usual meaning. Calculate γ in terms of λ so that $[A, A^\dagger] = 1$
- (b) Evaluate $\langle \psi | H | \psi \rangle$ where $\psi = \left(\frac{mw}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{mw}{2\hbar} x^2\right)$ and $H' = ve^{-\beta x^2}$

Paper 7 of 15 — Phy 311 Exam PQ and Ans

Level 300 • First Semester • PastQuestion

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Paper 8 of 15 — Phy 311 Test Qst and Ans

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EF

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS >

SECOND SEMESTER EXAMINATION. ‘

Course Code: PHY 311

Course Title: Quantum Mechanics

Course Unit: 3 Units

Session: 2019/2020

Time Allowed: 45 Minutes Date: 04-02-2021

Instruction: Answer al

e and in linear algebra. (3 m

fe 1. (a) Define an operator of quantum theory in the Hilbert space

\ matrix. (3 marks)

arks)

(b) Write a short note on Hermitian operator and Hermitian Mm

| Write a short note on expectation values. (4 marks) 2

(3) $\Psi(x) = \sin \frac{\pi x}{a}$ for $0 < x < a$ find $\langle x \rangle$ (6 marks) 4

Normalize the wavefunction

$\Psi(x) = \sin \frac{\pi x}{a}$ (4 marks)

ue(c)

e average value of $\langle P_x \rangle$.

$\Psi(x) = \sin \frac{\pi x}{a}$ considering the conditions $\Psi(0) = \Psi(a) = 0$

ve (b)

gand $\langle x \rangle = \frac{a}{2}$

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Paper 9 of 15 — Phy 311 Exam PQ 2

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN



DEPARTMENT OF PHYSICS

Course title: PHY 311

Course code: Quantum Mechanics I

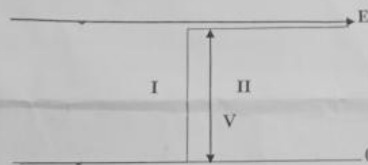
Time allowed: 3 hrs

Unit: 3 units

Date: 26/02/2021: 8.00 AM

Instructions: Answer any four questions

- (1a). List and explain with equations, the three pictures of quantum mechanics. (7½ marks)
- (1b). Show that $\phi = \begin{pmatrix} e^{-i\omega t/2} \\ e^{i\omega t/2} \end{pmatrix}$ satisfies the Schrodinger wave equation with $H = \omega S_z$. (4 marks)
- (1c). Obtain the SWE for a charged particle in an electromagnetic field characterized by the vector potential $\vec{A}$, and the scalar potential ϕ whose classical Hamiltonian is given by (6 marks)
- $$H = \frac{1}{2m} (\vec{p} - q\vec{A})^2 + q\phi$$
- (2). Consider a particle of mass m , and Energy E in the following potential step $V(x)$



$$V(x) = V_1 \quad \forall \quad -\infty < x < 0$$

$$= V_2 \quad \forall \quad 0 < x < \infty$$

Where I is the incident region and II is the transmitted region (i). Find the amplitudes at the boundary $x=0$ when $E > V$ (8½ marks) (ii). Find the reflection and transmission coefficients when $E > V$ (9 marks) Using the time independent Schrodinger wave equation.

$$\left[\frac{-\hbar^2 d^2}{2mdx^2} + V(x) \right] \psi(x) = E\psi(x)$$

- (3). A particle of mass m moves in a potential given by $V(x, y, z) = \frac{k}{2}(x^2 + y^2 + z^2) = \frac{kr^2}{2}$
- (a). Write down the time-independent Schrodinger equation for this system. (1 mark)
- (b). Make the substitution $\psi(x, y, z) = x(x)Y(y)Z(z)$ and separate the variables for this system (5 marks)
- (c). What are the solutions to the resulting equations for $X(x)Y(y)Z(z)$ (2½ marks)
- (d). What is the general expression for the quantized energy levels of this system, in terms of the quantum number n_x, n_y and n_z which corresponds to $X(x)Y(y)Z(z)$ (1 mark)
- (e). What is the degree of degeneracy of a state of energy $E = 5.5\hbar \sqrt{\frac{k}{m}}$ for this system? (4 marks)
- (f). An alternative solution may be the substitution, what are the solutions for $\psi(r, \theta, \phi) = F(r) G(\theta, \phi)$. In this substitution, what are the solutions for $G(\theta, \phi)$. (1 mark)
- (g). Write down the differential equation for $F(r)$ which is obtained when the substitution $\psi(r, \theta, \phi) = F(r) G(\theta, \phi)$ is made. Do not solve this equation. (3 marks)

- (4). For a 1-D Simple harmonic oscillator where $f(x) = -kx$, $V = \frac{1}{2}kx^2$. If the One-dimensional Schrodinger time-independent wave equation for the problem is given by $\left[\frac{-\hbar^2}{2m}\frac{d^2}{dx^2} + V\right]\psi = E\psi$ with the transformations $k = m\omega^2$, $y = \left(\frac{m\omega}{\hbar}\right)^{1/2}x = \beta x$, $\lambda = \frac{2E}{\hbar\omega}$, assuming a solution $\psi = H(y)e^{-1/2y^2}$, derive the ground state energy $E = \frac{\hbar\omega}{2}(2N+1)$ (17½ marks)

(5a). List five characteristic properties of the wave function of the quantum theory. (5 marks)

(5b). Normalize the following wave functions. (i). $\psi = xe^{-\alpha x^2}$ (2 marks)

(ii). $\psi = Ax^2$ where $-b/2 < x < b/2$ (2 marks)

(iii). $\psi(x) = A\sqrt{x} \cos\left(\frac{\pi x}{2b}\right)$ $0 < x < b$ (4½ marks)

(iv). $\psi = Ax(a-x)$ $0 \leq x < a$; $0 \leq x \leq a/b$ (4 marks)

(6a) (i). Define an operator in the Hilbert space and in linear algebra. (1½ marks)

(ii). Write a short note on Hermitian operator and Hermitian matrix (1½ marks)

(iii). Write the relation and the general solution for the Schrodinger equation in spherical coordinates. (1½ marks)

(iv). Write a short note on expectation values (1½ marks)

(6b). The normalized function of a particle is given by $\int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$, if the particle is found between $0 \leq x \leq L$, find the expectation value of the momentum P_x and position X . (6½ marks)

(6c). Obtain the expectation value for the momentum of a particle whose wave function is given by $\psi_1(x) = \frac{1}{\sqrt{b}} \exp(ikx)$ if $0 \leq x \leq b$ (5 marks)

Time Independent $\rightarrow -\hbar^2 \frac{d^2}{dx^2} \psi(x) + V(x) \psi(x) = E \psi(x)$

Time Independent $\rightarrow -\hbar^2 \frac{d^2}{dx^2} \psi(x,t) + V(x) \psi(x,t) = -\hbar^2 \frac{d}{dt} \psi(x,t)$

Paper 10 of 15 — Phy 311 Test PQ

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF PHYSICS
SECOND SEMESTER EXAMINATION.

Course Code: PHY 311

Course Title: Quantum Mechanics

Course Unit: 3 Units

Session: 2019/2020

Time Allowed: 45 Minutes Date: 04-02-2024

Instruction: Answer all questions

ear algebra. (3 marks)

he Hilbert space and in lin

ae 1. (a) Define an operator of quantum theory in tl

(b) Write a short note on Hermitian operator and Hermitian matrix. (3 marks)

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lues. (4 marks) Cc

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u(c) Write a short note on expectation va

x) (6 marks) @

oe 2. (a) For $W(x) = a \sin \frac{\pi x}{a}$ find the average

ve (b) Normalize the wavefunction $\psi = A \sin \frac{\pi x}{a}$ considering the c

qand $0 < x < a/3$ (4marks)

e value of $\langle P_x \rangle$.

onditions $0 < x < a$

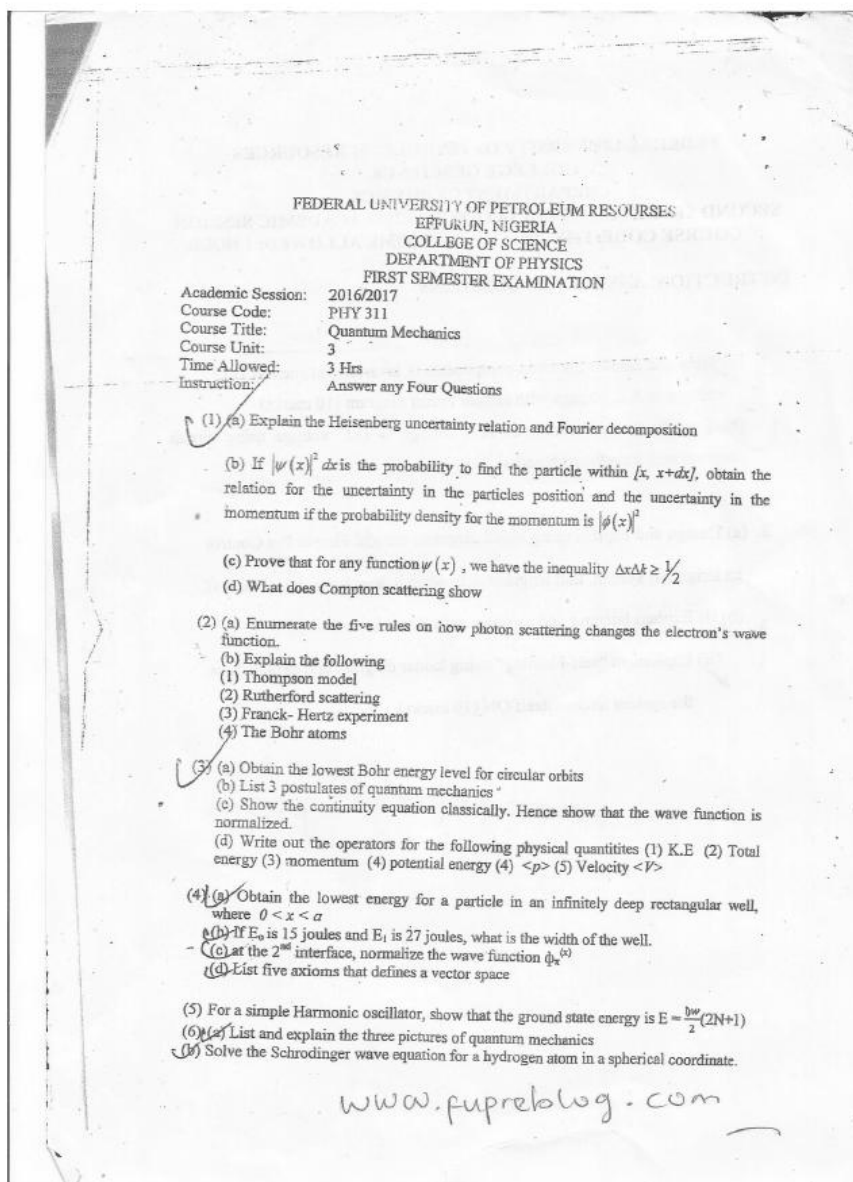
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Paper 11 of 15 — PHY 311 2016 2017

Level 300 • First Semester • PastQuestion

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Paper 12 of 15 — PHY 311 2017 2018

Level 300 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES |

EFFURUN, NIGERIA |

COLLEGE OF SCIENCE :

DEPARTMENT OF PHYSICS |

FIRST SEMESTER EXAMINATION |

Academic Session: 2017/2018

Course Code: PHY 317 |

Course Title: Quantum Mechanics ee |

Course Unit: 3 LWW. pupreblag Con

Time Allowed: 2hrs 30mins

Instruction: Answer any Four Questions

1a)

1b)

1c)

1d)

2)

4

3a)

i)

ii)

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Obtain the Schrodinger wave equation for a charged particle in an electromagnetic ee characterized by the vector potential A and the scalar potential ϕ whose clas

sical

Hamiltonian is given by

Dale a |

$H = -\frac{1}{2m}(\mathbf{p} - q\mathbf{A})^2 + q\phi$ |

2m |

The Hamiltonian of an electron situated in a magnetic field of induction $\mathbf{B} = (0, 0, B)$, is given by $H = \frac{1}{2m}(\mathbf{p} - q\mathbf{A})^2 + q\phi$ Obtain the equation of motion for (S_x) , (S_y) , and (S_z) and comment on m : |

your results.

Show that $[L_x, L_y] = \hbar L_z$, CF

If the energy difference in an infinitely deep potential well is $Z_p = 5J$ and $E_1 = 1$ ree the width of the potential well.

For a particle scattered from a potential step, obtain the relations for the reflection, and transmission coefficients when $E > 0$ |

A particle moving along the x-axis within the limits $(0, a)$ is describes by the wavefunction

$$\psi = \sin \frac{\pi x}{a}$$

Determine the normalizing factor and mean position of the particle.

Determine the uncertainty Δx in the position of the particle. |

A particle trapped in a region may be described by the wavefunction $\psi = A x(a-x)$

within the interval; $0 < x < a$ Where A is

the normalizing factor. Determine A and

the probability of finding the particle within the interval $Chis$

2

e

Show that. $\psi = [\text{wwf}]$ Satisfies the Schrodinger wave equation with $H = \frac{p^2}{2m} - V$ |

: eh) :

» 5 |

List and describe the three (3) pictures of quantum mechanics. _

4b)

4c)

4d)

5a)

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6c)

5 |

For $y, (x) = f_2 s_0(22)$ with $0 < x < L$, find the average value of $P-X$. That is $(P, -x)$

Ya

Evaluate $\langle w/H'/y \rangle$ for a simple harmonic oscillator, where $y = \exp(-x^2/2a^2)$

7a 2h

and $H' = -\frac{1}{2}kx^2$

A system is in a state described by the wave function $\psi =$

$\exp(-\sqrt{2}kx/a)$ Determine

nit =

the probability that a measurement of its energy will give a result corresponding to the

YY,

energy Eigenfunction, $\psi = A \exp(-f \cdot x/a)$; |

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a

A particle confined to a cubical box of volume V is found to be in a state given by the wavefunction $|\psi\rangle = Axyz$

Obtain the normalizing constant.

Determine the probability of finding the particle in the ground state of the box.

For a particle scattered from a potential step at $E > V$, find the percentage transmission and reflection if

$E = 6\text{eV}$, $V = 2\text{eV}$

$E = 3.24\text{eV}$, $V = 2\text{eV}$

Determine $x = y$ for $T = 75\%$ Ww uobo- pupre ble

Lr

$y \ll \text{CBM}$

d

Normalize the wavefunction $|\psi\rangle = | -3$

243i

The transmission coefficient in the scattering from a potential barrier is given as

$= i$

V Sink? 2m

Si OE | core == " (-V)

$A \exp(E-V)$;

If the width of the potential barrier is $a = 0.280\text{nm}$, the energy and potential are 5eV and 2eV respectively, determine T

If the energy and potential are 4eV and 6eV respectively for a reflection coefficient of $R = 90\%$, find k and a

$1/3 \rightarrow A \setminus$

$1/i$

$2W^2 \Rightarrow 7-1$

$A = | -WE 0.18 | \text{pyqe} | ?$

: | NO | 2

1) 0

Determine the expectation value of A in each of the state y , and y .

Determine the average of x' for a simple harmonic oscillator which is in a state define by the wavefunction $y/$ is @

2

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"3? \$

Paper 13 of 15 — PHY 311 2015 2016

Level 300 • First Semester • PastQuestion

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— UNIVERSITY OF PETROLEUM RESOURCES

- EFFURUN, NIGERIA.

COLLEGE OF SCIENCE

DEPARTMENT OF PHYSICS

FIRST SEMESTER EXAMINATION 2015/2016 ACADEMIC SESSION

COURSE CODE: PHY 311 TIME ALLOWED: 2:30 Hrs

COURSE TITLE: QUANTUM MECHANICS I COURSE UNIT: 3

INSTRUCTION: ANSWER ANY FOUR QUESTIONS

1a.

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2a.

(b)

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4b.

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Ses

1)

List five characteristics of a wavefunction

For 2 particle in the state given by $u = Ax'$, $0 < x < a$, determine the normalization constant A and the probability that the measurement will yield an eigenvalue

corresponding to the normalized eigenfunction, $y = 2 \sin$

a -A~—— ~

What is the De Broglie wavelength of an electron whose kinetic energy is 100eV

$6.626 \times 10^{-34} \text{ Js}$, $m_e = 9.109 \times 10^{-31} \text{ Kg}$). ' , caval

Using the uncertainty principle, estimate the ground state energy of an electron in the hydrogen atom ?

Obtain the general solution for a particle in an infinitely deep potential well.

Prove that $\langle \hat{B} \rangle = 2 \langle \hat{A} \rangle \langle \hat{I} \rangle$

Explain; prove

1. The correspondence principle

2. The Heisenberg uncertainty principle

3. The complementarity principle

Using the time dependent Schrodinger wave equation, show that $\psi(r,t) = \psi(r)e^{-iEt/\hbar}$ -
_iet

If $w(r,t) = \psi(r)e^{-iEt/\hbar}$ * prove that it satisfies the mass conservation equation

A system is in a state described by the wave function $\psi(r,t) = \psi(r)e^{-iEt/\hbar}$.

ei

Determine the probability that a measurement of its energy will give a result

22

: a? 1

corresponding to the energy eigenfunction $\psi_n = A e^{-n\pi x/a}$ =

2

An electron situated in a magnetic field of induction $B = (0, 0, B)$ have a Hamiltonian

"given by $H = \frac{p^2}{2m} - \mu_B B \sigma_z$. Obtain the equation of motion for $\langle \sigma_x \rangle$, $\langle \sigma_y \rangle$ and $\langle \sigma_z \rangle$

m

List and describe the three pictures of quantum mechanics.

Given that $A = (P - iA_x) \psi = \psi$ and $A_t = (P \sigma_x) \psi$ where σ_x, σ_y are constants and all

other quantities have their usual meaning. Express ψ in terms of A and σ_z . that

$[\sigma_x, \sigma_y] = 2i\sigma_z$

Evaluate $\langle \psi | H | \psi \rangle$ where $\psi = \frac{1}{\sqrt{2}} (e^{-i\theta} \psi_1 + e^{i\theta} \psi_2)$ and $H = \frac{p^2}{2m} + V(x)$ i = e

Paper 14 of 15 — PHY 311 2017 2018 TEST

Level 300 • First Semester • PastQuestion

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Paper 15 of 15 — PHY 311 2012 2013 QUANTUM MECHANICS 1

Level 300 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES.

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COLLEGE OF SCIENCE

DEPARTMENT OF PAYSICS

FIRST SEMESTER EXAMINATION » 2012/2913 ACADEMIC SESSION

COURSE CODE: PHY 311 TYME ALLOWED: 3 LIRS

COURSE TITLE: QUANTUM MECHANICS. CURSE UNIT: 3 Y

INSTRUCTION: ANSWER ANY FOUR QUESTION? i if :

on Explain ;

an ve 2 Qe |. The correspondence principle

I< "ee 2. The Heisenberg uncertainty principle: ~

Ye ? 4 3. The complementarity principle

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$y = Ax(a-x)$ for $0 < x < a$. a

Determine 4 and the probability of. finding the particle within theinter:al(. -'3). (se

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Starting from the time dependent Schrodinger wave equation, obtain the exact a

solution for a free particle. : 5

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am. if wie $O=\phi_9)e''$, prove that it satisfies the mass conservation equation $V \text{ spat}$

5a. for a potential barrier of infinite-width, obtain the pened solution for scattering from |

lo a potential step of height V_0 in the regions $x < 0$ and $x > 0$ 8

Kee If $y = AVX \cos =$) in the region $0 < x < b$, Aiud the constant of sormAlizaton |

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Ga. a ee 'x an infinite potential well $0 < x < q$ is described bythe: sbavefantion 4

ae = 3x. Determine the = 'obability that it is in the,third: excited) state,

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~©) Derive the relation $\frac{dy}{dx}$ for the one dimensional separation of variables
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